WORKSHEET

Chapter 2: Expanding algebra

Part A

Expand each of the following:

1 $2(a + 3) =$

2 $3(a + 5) =$

3 $3(p - 2) =$

4 $5(5 + 2p) =$

5 $3(3m + 6) =$

6 $-2(2p - 1) =$

7 $3(a - 4) =$

8 $5(1 - 2p) =$

9 $-4(3p + 4) =$

10 $6(2x + 4) =$

11 $-m(m - 1) =$

12 $2p(3p^2 + 5) =$

13 $3d(2a + 5d) =$

14 $-2(2 + 2x) =$

15 $-3p^2(4p - 2) =$

16 $6(-3p + 2) =$

17 $-5(2y - 3) =$

18 $-n(2n^2 - 8n) =$

19 $2x^3(3x^4 - 4x) =$

20 $abc(2abc - 1) =$

Part B

Expand and simplify these where possible.

1 $2(a + 3) + 5 =$

2 $3(2y - 5) - 2y =$

3 $3 + 5(4p + 2) =$

4 $7 - 2(3m + 6) =$

5 $2(m + 1) + 3(2m - 5) =$

6 $6(p - 2) + 3(p + 4) =$

7 $a(bc - 2) - b(ac - 3) =$

8 $-3(x - 4) - 2x + 10 =$

9 $6 - (2p + 3) + 5(p - 1) =$

10 $3x - (2 - 3x) + 2 =$

11 $7p(1 - 2p) - 3p(2p - 4) =$

12 $2a(a - 3p) + a(1 - 4a) =$

13 $2(3a - 2 + b) - 2(3a + 4 + b) =$

14 $6(2a^2 + 3) + 2a(a + 1) - 3(a^2 + 4) =$

15 $2(7r - 5) - 2(3r - 2) - 5(1 - 2r) + r =$

16 $2m^3n^4(m - n) - 3mn(m^3n^3 - 5m^2n^4) =$

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Answers

Part A

1 $2a + 6$
2 $3a + 15$
3 $3p - 6$
4 $25 + 10p$
5 $9m + 18$
6 $-4p + 2$
7 $3a - 12$
8 $5 - 10p$
9 $-12p - 16$
10 $12x + 24$
11 $-m^2 + m$
12 $6p^3 + 10p$
13 $6ad + 15d^2$
14 $-4 - 4x$
15 $-12p^3 + 6p^2$
16 $-18p + 12$
17 $-10y + 15$
18 $-2n^3 + 8n^2$
19 $6x^7 - 8x^4$
20 $2a^2b^2c^2 - abc$

Part B

1 $2a + 11$
2 $4y - 15$
3 $20p + 13$
4 $-6m - 5$
5 $8m - 13$
6 $9p$
7 $3b - 2a$
8 $-5x + 22$
9 $3p - 2$
10 $6x$
11 $-20p^2 + 19p$
12 $-2a^2 - 6ap + a$
13 -12
14 $11a^2 + 2a + 6$
15 $19r - 11$
16 $13m^3n^5 - m^4n^4$

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